On de Bruijn arrays

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1. INTRODUCTION

The topic of a periodic binary sequence of length 2^n in which each of the different n-tuples appears exactly once existed in the mathematical literature since 1894, but continued to be rediscovered from time to time [2]. It became familiar to mathematicians through the works of de Bruijn and Good (independently) in 1946 [1, 3]. Following the crowd we shall call such sequences de Bruijn sequences of span n. In this paper we investigate a generalization of this object to the two-dimensional case.

Definition 1.1 An $(r, s; m, n)$-de Bruijn array (or from now on an $(r, s; m, n)$-array for short) is a periodic $r \times s$ binary array (with $m \leq r$, $n \leq s$, $rs = 2^{mn}$) in which each of the different $m \times n$ binary matrices appears exactly once.

Remark 1.2 When $r = 1$ or $s = 1$, an $(r, s; m, n)$-array is a de Bruijn sequence of span n or m respectively. However, unlike its one-dimensional analogue, $rs = 2^{mn}$ is no longer sufficient to ensure the existence of an $(r, s; m, n)$-array. For instance, there is no $(2, 2^{2n-1}; 2, n)$-array since the $2 \times n$ array with all 0 is bound to appear more than once.

Remark 1.3 The case where only non-zero $m \times n$ matrices are considered has been treated extensively in [4, 7, 8, 9] and is a 2-dimensional generalization of the maximal length sequence. Another variation on such kinds of "window properties" is discussed in [10].

We ask: (1) For which r, s, m, n does there exist an $(r, s; m, n)$-array? (2) In case an $(r, s; m, n)$-array exists, how can we construct one? (3) How many $(r, s; m, n)$-arrays are there? We have not attempted question (3) at all, but shall offer some construction methods, thereby deriving partial answers to questions (1) and (2). In particular we obtain the following results.

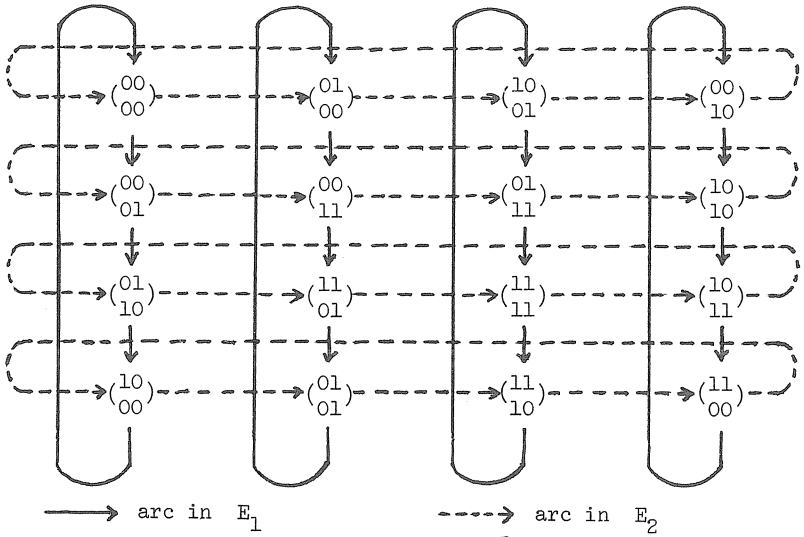
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Theorem 1.4 For any m, n, there are r, s such that an $(r, s; m, n)$-array exists.

Theorem 1.5 There exists an $(s, s; n, n)$-array if and only if n is even.

2. A GRAPH-THEORETIC FORMULATION

Analogous to the de Bruijn graph (also known as a Good diagram) G_n [5, section II] we introduce the digraph $G_{m,n}(V, E_1, E_2)$ where V is the set of all $m \times n$ binary matrices; E_1 is the set of all arcs joining M_1 to M_2 where the last $m-1$ rows of M_1 are the first $m-1$ rows of M_2; E_2 is the set of all arcs joining N_1 to N_2 where the last $n-1$ columns of N_1 are the first $n-1$ columns of N_2. An $(r, s; m, n)$-array corresponds to a subgraph of $G_{m,n}(V, E_1, E_2)$ which includes all vertices in V and which consists of s disjoint cycles of length r with arcs in E_1, and in which corresponding vertices are joined by arcs in E_2 to form r disjoint cycles of length s. The following is an example with $r = 4$, $s = 4$, $m = 2$, $n = 2$.



The corresponding $(4, 4; 2, 2)$-array is $A = \begin{bmatrix} 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 \end{bmatrix}$, which has been discussed in detail in [9].

3. EXISTENCE OF $(r, s; m, n)$-ARRAY

We can construct an $(2^m, 2^{m(n-1)}; m, n)$-array with $n \geq 3$ from a de Bruijn sequence of span m. The main idea is to place $2^{m(n-1)}$ copies of the given de Bruijn sequence as columns side by side, but each suitably shifted so that any consecutive $n-1$ columns have relative shifts $t_1, \ldots, t_{n-1}$ $(0 \leq t_i \leq 2^m-1)$ distinct (regarded as an $(n-1)$-tuple). This can be easily achieved by taking a de Bruijn sequence of span $n-1$ consisting of symbols chosen among $0, 1, 2, \ldots, 2^m-1$ (instead of just $0, 1$). Since the actual shifts are given by $w_0, w_1, w_2, \ldots$ where $w_0 = 0$, $w_i = t_0 + t_1 + \ldots + t_{i-1}$ for $i \geq 1$, we must also ensure that $w_0, w_1, w_2, \ldots$ is periodic of period $2^{m(n-1)}$. This is indeed the case as long as $n \geq 3$. For the detailed proof and an example of this construction, see [6, Theorem 2].

With this construction we can prove Theorem 1.4. For $m = 1$ or $n = 1$, a de Bruijn sequence is an answer. For $n \geq 3$, we have what we want from the discussion above. For $n < 3$ but $m \geq 3$, we first construct an $(2^n, 2^{n(m-1)}; n, m)$-array as above. Its transpose will be an $(2^{n(m-1)}, 2^n; m, n)$-array. The only remaining case of $m = n = 2$ is answered by the example given at the end of section 2.

4. A GENERALIZATION OF LEMPEL'S HOMOMORPHISM

For typographical convenience we shall denote by $(C_1 \ldots C_n)$ the matrix with columns $C_1, \ldots, C_n$; and by $(R_1^T \ldots R_m^T)^T$ the matrix with rows $R_1, \ldots, R_m$. We write $G_{m,n}$ for $G_{m,n}(V, E_1, E_2)$. Consider the mapping $D: G_{m+1,n} \to G_{m,n}$ given by

$$D(R_1^T \ldots R_{m+1}^T)^T = (S_1^T \ldots S_m^T)^T$$

where $S_i = R_i + R_{i+1}$ $(i = 1, 2, \ldots, m)$. It is easy to check that D is a graph homomorphism and that D is 2^n-to-one. The preimages of $(R_1^T \ldots R_m^T)^T$ are of the form $(R^T \; R^T+R_1^T \; R^T+R_1^T+R_2^T \; \ldots \; R^T+R_1^T + \ldots + R_m^T)^T$ where $R = (a_1 \ldots a_n)$ is an arbitrary binary n-tuple. For $n = 1$, this reduces to Lempel's homomorphism [5, section III].

Let $\mathcal{C}$ be a cycle of length r $(\geq m)$ in $G_{m,n}$ with arcs in E_1.

We can represent $\mathcal{C}$ as an $r \times n$ array $(R_0^T\ R_1^T \cdots R_{r-1}^T)^T$, meaning that the vertices on $\mathcal{C}$ are the $m \times n$ matrices $(R_i^T\ R_{i+1}^T \cdots R_{i+m-1}^T)^T$ $(i \equiv 0, 1, \ldots, r-1 \pmod r)$ in that order. It is not hard to convince oneself that the preimages of $\mathcal{C}$ consist of disjoint cycles in $G_{m+1,n}$ with arcs in E_1. Actually one can say much more.

(A) If $(\sigma_1 \cdots \sigma_n) = (0 \cdots 0)$ where σ_i is the column sum (mod 2) of the ith column of $\mathcal{C}$, then the preimages of $\mathcal{C}$ consist of 2^n disjoint cycles $\mathcal{C}_1, \ldots, \mathcal{C}_{2^n}$ in $G_{m+1,n}$ with arcs in E_1, each of length r. On each $\mathcal{C}_i$, exactly one vertex corresponds to the vertex $(A_1^T \cdots A_m^T)^T$, given by $(R^T\ R^T+A_1^T \cdots R^T+A_1^T + \cdots + A_m^T)^T$ where R is an arbitrary binary n-tuple.

(B) If $(\sigma_1 \cdots \sigma_n) \neq (0 \cdots 0)$, then the preimages of $\mathcal{C}$ consist of 2^{n-1} disjoint cycles $\mathcal{C}_1, \ldots, \mathcal{C}_{2^{n-1}}$ in $G_{m+1,n}$ with arcs in E_1, each of length $2r$. On each $\mathcal{C}_i$, exactly two vertices correspond to the vertex $(A_1^T \cdots A_m^T)^T$, given by $(R^T\ R^T+A_1^T \cdots R^T+A_1^T + \cdots + A_m^T)^T$ where R is an arbitrary binary n-tuple or its sum with $(\sigma_1 \cdots \sigma_n)$.

We shall omit the straightforward checking.

5. A CONSTRUCTION METHOD

We shall explain one method to obtain an $(r, 2^n s; m+1, n)$-array or an $(2r, 2^{n-1}s; m+1, n)$-array from an $(r, s; m, n)$-array A. Each of the $r \times n$ subarrays $\mathcal{A}_i$ $(i = 1, \ldots, s)$ of A can be regarded as a cycle of length r in $G_{m,n}$ with arcs in E_1 (as explained in section 4). These s cycles are disjoint since A is an $(r, s; m, n)$-array. The idea is to link up the preimages in $G_{m+1,n}$ of these cycles by arcs in E_2 in a suitable way. (See section 2).

Construction 5.1

Suppose $(\sigma_1 \cdots \sigma_s) = (0 \cdots 0)$ where σ_i is the column sum (mod 2) of the ith column of A, then by (A) in section 4 we know the preimages of each $\mathcal{A}_i$ $(i = 1, \ldots, s)$ consist of 2^n disjoint cycles $\mathcal{A}_{i,j}$ $(j = 0, 1, \ldots, 2^n-1)$ in $G_{m+1,n}$ with arcs in E_1, each of length r. Altogether we have $2^n s$ disjoint cycles, which we can arrange so that $\mathcal{A}_{i,j}$ contains the $(m+1) \times n$ matrix $A_{i,j}$ with its first row being the binary representation of j. It suffices then to find a cycle of length $2^n s$ with arcs in E_2 which link up

$$
\begin{array}{c}
A_1, \pi_1(0), A_2, \pi_2(0), \cdots, A_s, \pi_s(0), \\
A_1, \pi_1(1), A_2, \pi_2(1), \cdots, A_s, \pi_s(1), \\
A_1, \pi_1(2), A_2, \pi_2(2), \cdots, A_s, \pi_s(2), \\
\cdots\cdots\cdots\cdots\cdots \\
A_1, \pi_1(2^n-1), A_2, \pi_2(2^n-1), \cdots, A_s, \pi_s(2^n-1), A_1, \pi_1(0)
\end{array}
$$

in that order, where π_i is a permutation of $\{0, 1, \ldots, 2^n-1\}$. This is because corresponding matrices on these $2^n s$ cycles will be linked up by a cycle of length $2^n s$ with arcs in E_2 as well, thanks to the design of D and the fact that $\mathcal{A}_1, \ldots, \mathcal{A}_s$ are similarly linked. For the same reason, it suffices further to confine attention to only the first rows of these $A_{i,j}$'s. Thus, the problem reduces to find a binary sequence of period $2^n s$, say $b_1 b_2 \cdots b_{2^n s}$ such that for each i $(1 \leq i \leq s)$, the 2^n n-tuples $b_{i+ts} b_{i+1+ts} \cdots b_{i+n-1+ts}$ $(t = 0, 1, 2, \ldots, 2^n-1)$ are all distinct. This can be achieved as follows. Take a de Bruijn sequence of span n with n 0's at the beginning. Delete the first zero to obtain a sequence $c_1 c_2 \cdots c_{2^n-1}$. Define

$$
\begin{aligned}
&b_1 = \cdots = b_s = 0 \\
&b_{s+j+k(2^n-1)} = c_j \quad (j = 1, \ldots, 2^{n-1};\ k = 0, 1, \ldots, s-1).
\end{aligned}
$$

Since s is a power of 2, we know 2^n-1 and s are relatively prime, and hence for each i $(1 \leq i \leq s)$, the 2^n integers $i+ts$ $(t = 0, 1, \ldots, 2^n-1)$ are all distinct (mod 2^n-1), which means $b_1 \cdots b_{2^n s}$ is a solution to our problem, thanks to the property enjoyed by $c_1 \cdots c_{2^n-1}$.

Remark 5.2 What one actually does in Construction 5.1 is to write down $b_1 \cdots b_{2^n s}$ as the first row, then append successive rows by adding cumulatively the rows of $[A \vdots A \vdots \cdots \vdots A]$ (2^n copies) to the first row. Since there are $2^{n-1}s$ 1's in the sequence

$$b_1 \cdots b_{2^n s} = \underbrace{0 \cdots 0}_{s \text{ of them}} \underbrace{c_1 c_2 \cdots c_{2^n-1}}_{1st} \cdots\cdots \underbrace{c_1 c_2 \cdots c_{2^n-1}}_{sth}$$

and there are $2^n w$ 1's in each row of $[A \vdots A \vdots \cdots \vdots A]$ where w is the number of 1's in the corresponding row in A, we see that all row sums of the resultant array are even.

Remark 5.3 Suppose each column $(d_1 \ldots d_r)^T$ of A satisfies the property that $d_1 + d_3 + d_5 + \cdots + d_{r-1} \equiv 0 \pmod 2$, then all column sums of the resultant array are even. This is because each column of the resultant array is of the form $(d \;\; d+d_1 \;\; d+d_1+d_2 \cdots d+d_1 + \cdots + d_{r-1})^T$ where d is either 0 or 1. Adding up, we obtain $rd + (r-1)d_1 + (r-2)d_2 + \cdots + 2d_{r-2} + d_{r-1} \equiv d_1 + d_3 + d_5 + \cdots + d_{r-1} \equiv 0 \pmod 2$ since r (being a power of 2) is even.

Example 5.4 We start with the (4, 16; 2, 3)-array

$$A = \begin{bmatrix} 0&0&0&0&1&0&1&0&0&0&1&1&0&1&0&1 \\ 0&0&0&1&1&0&1&1&0&0&1&1&1&1&1&0 \\ 1&1&1&1&0&1&0&1&1&1&0&0&1&0&1&0 \\ 1&1&1&0&0&1&0&0&1&1&0&0&0&0&0&1 \end{bmatrix}$$

[6, Example to Theorem 2]. We construct the binary sequence of period 128 as required, viz.

$$\underbrace{00 \cdots 0}_{16 \text{ of them}} \underbrace{0011101}_{1st} \underbrace{0011101}_{2nd} \cdots \underbrace{0011101}_{16th}$$

and use it to obtain (see Remark 5.2) the (4, 128; 3, 3)-array

$$\begin{array}{c} (1)\;(2)\;(3)\;(4)\;\cdots\;(16)(17)(18)(19)\;\cdots\cdots\;(125)(126)(127)(128) \\ \begin{bmatrix} 0&0&0&0&\cdots&0&0&0&1&\cdots\cdots&1&1&0&1 \\ 0&0&0&0&\cdots&1&0&0&1&\cdots\cdots&1&0&0&0 \\ 0&0&0&1&\cdots&1&0&0&1&\cdots\cdots&0&1&1&0 \\ 1&1&1&0&\cdots&1&1&1&0&\cdots\cdots&1&1&0&0 \end{bmatrix} \end{array}$$

Construction 5.5

Suppose $(\sigma_1 \ldots \sigma_s) = (1 \ldots 1)$ where σ_i is the column sum (mod 2) of the ith column of A, then by (B) in section 4 we know that the preimages of each $\mathcal{A}_i$ $(i = 1, \ldots, s)$ consist of 2^{n-1} disjoint cycles $\mathcal{A}_{i,j}$ $(j = 0, 1, \ldots, 2^{n-1}-1)$ in $G_{m+1,n}$ with arcs in E_1, each of length $2r$. Altogether we have $2^{n-1}s$ disjoint cycles, which we can arrange so that $\mathcal{A}_{i,j}$ contains the $(m+1) \times n$ matrix $A_{i,j}$ with its first row being the binary representation of j (in which case $\mathcal{A}_{i,j}$ also contains the $(m+1) \times n$ matrix $A'_{i,j}$ whose rows are dual to those of $A_{i,j}$ in the sense that they differ by $(1 \cdots 1)$). Similar reasoning as before reduces the problem to finding a binary sequence of period $2^{n-1}s$, say $b_1 \cdots b_{2^{n-1}s}$ such that for each i $(1 \leq i \leq s)$, the 2^{n-1} n-tuples $b_{i+ts} b_{i+1+ts} \cdots b_{i+n-1+ts}$ $(t = 0, 1, \ldots, 2^{n-1}-1)$ are all distinct and no two are dual to each other. This can be achieved as follows. For $n = 1$, define $b_i = 0$ $(i = 1, \ldots, 2^{n-1}s)$. For $n = 2$, define $b_1 = \cdots = b_s = 0$, $b_{s+2k-1} = 0$, $b_{s+2k} = 1$ $(k = 1, \ldots, s(2^{n-1}-1)/2)$. For $n \geq 3$, define $b_1 = \cdots = b_s = 0$, $b_{s+j+k(2^{n-1}-1)} = c_j$ $(j = 1, \ldots, 2^{n-1}-1;\; k = 0, 1, \ldots, s-1)$ where $c_j = a_0 + a_1 + \cdots + a_{j-1}$ with $a_0 a_1 \cdots a_{2^{n-1}-1}$ being a de Bruijn sequence of span $n-1$ with $n-1$ 0's at the beginning. The argument is basically the same as before except for one technical variation whose detail is provided by [5, Theorem 5].

Example 5.6 Let A be the (4, 4; 2, 2)-array given at the end of section 2. We first construct the binary sequence of period 8 as required, viz. 00000101, and use it to obtain the (8, 8; 3, 2)-array

$$\begin{bmatrix} 0&0&0&0&0&1&0&1 \\ 0&0&1&0&0&1&1&1 \\ 0&0&1&1&0&1&1&0 \\ 0&1&0&0&0&0&0&1 \\ 1&1&1&1&1&0&1&0 \\ 1&1&0&1&1&0&0&0 \\ 1&1&0&0&1&0&0&1 \\ 1&0&1&1&1&1&1&0 \end{bmatrix}$$

6. ON SQUARE DE BRUIJN ARRAYS

The next result guarantees repeated applications of Construction 5.1 to certain de Bruijn arrays.

Theorem 6.1 If A is an (s, s; n, n)-array with all column sums even, then there is a (t, t; n+2, n+2)-array with all column sums even. (Note that $s = 2^{n^2/2}$, so n must be even and $s \geq 4$.)

Proof. Apply Construction 5.1 to A to obtain an $(s, 2^n s; n+1, n)$-array A_1 with all row sums even (see Remark 5.2). A_1^T is then an $(2^n s, s; n, n+1)$-array with all column sums even. Apply Construction 5.1 to A_1^T to obtain an $(2^n s, 2^{n+1} s; n+1, n+1)$-array A_2. Note that all column sums of A_2 are even because columns of A_1^T (= rows of A_1) are of the form

$$(b_1 + r_1 \quad b_2 + r_2 \quad \cdots \quad b_{2^n s} + r_{2^n s})^T$$

where $b_1 \cdots b_{2^n s}$ is of the type described in Construction 5.1 and $r_1 \cdots r_{2^n s}$ is of period s, so that $(b_1+r_1)+(b_3+r_3)+\cdots+(b_{2^n s-1}+r_{2^n s-1}) \equiv b_1+b_3+\cdots+b_{2^n s-1} \equiv (s/2)(2^{n-1}) \equiv 0 \pmod 2$ (see Remark 5.3). We can thus apply Construction 5.1 to A_2 to obtain an $(2^n s, 2^{2n+2} s; n+2, n+1)$-array A_3 with all row sums even (see Remark 5.2). A_3^T is then an $(2^{2n+2} s, 2^n s; n+1, n+2)$-array with all column sums even. Apply Construction 5.1 to A_3^T to obtain an $(2^{2n+2} s, 2^{2n+2} s; n+2, n+2)$-array A_4. Note that all column sums of A_4 are even (see Remark 5.3 and the argument offered just a few lines above).

To establish Theorem 1.5 from Theorem 6.1 we need only find an $(256, 256; 4, 4)$-array with all column sums even. We start with the following $(4, 16; 3, 2)$-array [6, Example to Theorem 3]

$$A = \begin{bmatrix} 0&0&1&1&0&0&1&0&0&0&1&1&0&1&1&1 \\ 0&0&1&1&0&0&0&1&0&1&1&0&0&0&0&1 \\ 0&0&1&1&0&1&1&1&0&0&1&1&0&0&1&0 \\ 1&1&0&0&1&0&1&1&1&0&0&1&1&0&1&1 \end{bmatrix}$$

A^T is an $(16, 4; 2, 3)$-array with all column sums even, so it yields an $(16, 32, 3, 3)$-array A_1 by Construction 5.1. The column sums of A_1 are all even since the condition in Remark 5.3 is satisfied by A^T. We can thus apply Construction 5.1 to A_1 to obtain an $(16, 256; 4, 3)$-array A_2 with all row sums even (see Remark 5.2). A_2^T is then an $(256, 16, 3, 4)$-array with all column sums even. Apply Construction 5.1 to A_2^T to obtain an $(256, 256; 4, 4)$-array A_3. The column sums of A_3 are all even (see Remark 5.3 and the argument offered in the proof of Theorem 6.1).

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